GCODECrunch — Development & Performance Report

Project: GCODECrunch G-code travel optimiser **Test set:** `Aircraft/Planks/Gruel/Gruel EMX07/GCODE`
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Report status: running document — appended after each experiment iteration.

1. Executive summary

GCODECrunch reorders extrusion segments within each print layer to cut non-extruding travel. The work has progressed through three tool generations, six ordering strategies, and an inter-layer coupling pass. This report tracks each step and its measured effect on travel distance across a fixed test set of eight production wing/plank G-code files.

Across the eight files (3 246 703 mm of baseline travel), the optimiser currently removes **63.6 %** of non-extruding travel:

Strategy	Total travel (mm)	Saved vs baseline	Wall time (s)
baseline (raw source)	3 246 702.7	—	—
`verbatim`	3 047 523.3	6.1 %	18.7
`greedy`	1 365 715.1	57.9 %	19.2
`greedy+2opt` (coupled)	1 197 764.2	63.1 %	56.7
`greedy+2opt+oropt` (uncoupled)	1 213 959.5	62.6 %	30.5
`greedy+2opt+oropt` (coupled)	1 182 242.9	63.6 %	61.7
`auto` (adopt best)	1 182 242.9	63.6 %	163.6

The extruding distance is preserved to within < 0.001 % (correctness invariant). Wall times are for the full eight-file set on the current machine.

Three findings stand out. First, the greedy seed already captures most of the available saving; 2-opt and Or-opt add successive ~3 % and ~0.9 %. Second, inter-layer coupling — steering each layer's exit toward the next layer's entry — adds a further **−2.6 %** on top of Or-opt (and is worth even more to plain 2-opt: `greedy+2opt` coupled beats uncoupled `greedy+2opt+oropt`). Third, work is approaching completion: over 82 % of the remaining travel already sits in layers that are fully reorderable, so further gains must come from better solutions to the same per-layer problem (or below whole-segment granularity), not from reordering more layers.

A subtle bug was found and fixed during this work: Or-opt's insertion-cost model used the segment's source orientation rather than its current orientation, producing a handful of moves that *increased* travel (the bug is documented in §6, Iteration 3).

2. Objective

1. Reduce non-extruding (travel) distance in sliced G-code without changing the deposited geometry.
2. Keep reversal (boustrophedon) available at all times.

3. Make the segment-ordering strategy selectable, including an automatic "adopt the best solution" mode.
4. Keep right-hand and left-hand wing parts symmetric — mirrored pairs must end up with equal travel.

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3. Test set and method

3.1 Files

Top-level `*.gcode` in the test folder (sub-folders `OPT/` and `Retired/` excluded):

File	Side	Pair key
`Elevon Horn x 2.gcode`	(single)	—
`Gruel Wing Plates.gcode`	(single)	—
`LOuterGruel VI.gcode` / `ROuterGruel VI.gcode`	L / R	`OuterGruel VI`
`LR Gruel VI.gcode` / `RR Gruel VI.gcode`	L / R	`R Gruel VI`
`LT Gruel VI.gcode` / `RT Gruel VI.gcode`	L / R	`T Gruel VI`

The `L`/`R` prefix denotes left/right hand; a mirrored pair is expected to exhibit identical travel statistics.

3.2 Metric

`travel_dist` from `gc_engine.analyze_path` / `analyze_lines`: total length of non-extruding XY/Z moves, in millimetres, parsed with the same state machine the optimiser uses. Lower is better. `total_dist` and extruding distance are recorded for context. The extruding distance must be invariant across strategies (a correctness invariant).

3.3 Harness

`bench_gcode.py` (project root) runs the baseline plus each requested strategy over every test file and emits:

- a Markdown results block (pasted here verbatim), and
- `bench_results.csv` (raw per-file/per-strategy metrics).

Run:

```
cd "/home/gke/Documents/Flight/3D Projects" && python3 bench_gcode.py
```

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4. Track of work

Point	Artifact	What changed
v1	`GCODECrunch_v1.py`	First self-contained GUI: folder/file select, NN reorder, temp gradient.
v2	`GCODECrunch_v2.py`	Refactor of the NN GUI (input validation, progress, safer I/O).
v3	`GCODECrunch.py` + `gc_engine.py`	Engine split out as a clean library/CLI; GUI is a thin wrapper.
v3.1	+ temperature gradient	Layer-count-based LW gradient; `enable` checkbox; `M104`/`M109` rewrite only.
v3.2	+ output naming	`{timestamp}_{safe_name}.gcode`; versioned copies; OctoPrint-friendly.

Point	Artifact	What changed
v3.3	+ pluggable orderings	`order` option: `verbatim` / `greedy` / `greedy+2opt` / `auto`.
v3.4	+ 2-opt & auto	Open-path 2-opt local search; `auto` scores candidates and adopts the best.
v3.5	+ Or-opt	Single-segment relocation local search (optionally reversed) run after 2-opt; added to `auto`'s candidate pool.
v3.6	+ inter-layer coupling	`couple_layers` option: re-sweep each layer so its exit steers toward the next layer's entry. Best sweep kept; strictly non-worsening.

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5. Strategy catalogue

Strategy	Description	Cost
`verbatim`	Emit segments in source order (no reordering). Control case; still re-emits connectors.	O(n)
`greedy`	Greedy nearest-neighbour from the current printer position, optional reversal. The classic v1/v2 behaviour.	O(n²)/layer
`greedy+2opt`	Greedy seed followed by open-path 2-opt local search (segment count ≤ `two_opt_max_segments`).	O(n²)–O(n³)/layer
`greedy+2opt+oropt`	As above, then Or-opt single-segment relocation (`or_opt_max_segments`, `or_opt_max_moves`).	O(n²)–O(n³)/layer
`auto`	Builds `greedy`, `greedy+2opt` and `greedy+2opt+oropt`, scores all with `analyze_lines`, adopts the lowest travel. Reports which won.	sum of all

Baseline (raw source file, no re-emit) is also reported for reference. `verbatim` differs from baseline only because connectors are regenerated on re-emit.

5.1 Inter-layer coupling (`couple_layers`)

Orthogonal to the ordering strategy. When `couple_layers ≥ 1`, `_nn_build` runs a first pass, records each layer's chosen entry point, then re-sweeps the whole file several times passing each layer a target: the entry point of the following layer. The local searches then optimise the path as *start* → … → *target*, so a layer is encouraged to finish close to where the next one begins. The connector between layers is the regenerated travel into the next layer's first segment, whose length is exactly `distance(exit_prev, entry_next)`.

Because `_nn_build` scores every sweep with `analyze_lines` and keeps the lowest, a non-zero `couple_layers` can only improve or equal the travel; sweeps are separate from the ordering strategy. Cost is roughly `(1 + couple_layers) ×` the uncoupled time. Default is `1`.

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6. Experiment log

Iteration 1 — establish the baseline and per-strategy envelope

Hypothesis: `greedy` improves materially over baseline; `greedy+2opt` improves further on layers small enough to enter the 2-opt pass; `auto` never loses to `greedy` and equals `greedy+2opt` where it helps.

Results (travel mm; wall time is per-file, full-set totals in §1):

File	Lines	Layers	Segs	Baseline	verbatim	greedy	greedy+2opt	auto
Elevon Horn x 2	6 245	16	174	3 000.4	2 772.0	2 350.9	2 253.9	2 253.9
Gruel Wing Plates	22 730	25	852	9 914.5	7 347.9	4 550.5	4 196.2	4 196.2
LOuterGruel VI	121 759	488	3 813	154 557.9	141 374.4	57 185.5	56 936.6	56 936.6
LR Gruel VI	562 531	520	13 705	443 449.0	412 196.7	176 634.3	145 831.1	145 831.1
LT Gruel VI	1 041 852	2 206	22 067	871 749.7	815 256.3	450 211.2	411 455.6	411 455.6
ROuterGruel VI	120 928	488	3 810	153 932.7	141 362.2	57 533.6	56 966.4	56 966.4
RR Gruel VI	629 242	612	15 719	562 574.2	530 064.0	162 974.4	139 340.5	139 340.5
RT Gruel VI	1 039 161	2 206	22 069	1 047 524.3	997 149.8	454 274.7	408 271.2	408 271.2

Totals: baseline 3 246 702.7; `verbatim` 3 047 523.3 (−6.13 %); `greedy` 1 365 715.1 (−57.94 %); `greedy+2opt` 1 225 251.4 (−62.26 %); `auto` equal to `greedy+2opt` (−62.26 %). 2-opt beat greedy by a further 10.3 %.

The extruding-distance invariant held: deltas were < 0.001 % (largest ≈ +2.58 mm over a ~326 000 mm extrusion total).

Commentary. The dominant win is the greedy nearest-neighbour seed; 2-opt adds a solid second helping. `auto` chose `greedy+2opt` on every file — as expected, since 2-opt never makes a layer worse. This iteration left the question: how much travel remains, and can a stronger local search reach it?

Iteration 2 — is the optimiser symmetric under mirroring?

Hypothesis: if the segment-ordering algorithm is mirror-invariant, then any left/right travel difference is inherited from the input files, not introduced by the optimiser.

Method. Take `LT Gruel VI`, mirror it about its own mid-plane (X span 0…967, k = 967), and run every strategy on both. A mirror-invariant optimiser yields a constant offset between original and mirror for every strategy.

Results (travel mm):

Strategy	original	mirrored	delta
raw baseline	871 749.7	872 690.8	−941.1
`verbatim`	815 256.3	816 197.4	−941.1
`greedy`	450 211.2	451 152.4	−941.1
`greedy+2opt`	411 455.6	412 396.7	−941.1

The delta is **identical (−941.1 mm) for all strategies.**

Commentary. The optimiser adds *zero* asymmetry: mirroring the input moves the result by exactly the same amount the raw file moves. Therefore any residual L−R travel gap is a property of the source geometry, not of the reordering. This redirected the symmetry investigation to comparing the actual input files.

Iteration 3 — Or-opt (single-segment relocation)

Motivation. A per-layer travel breakdown showed the remaining travel is not in unreachable corners: it sits in layers that *are* being reordered.

File	start-up block	reordered layers (≥2 segs)	verbatim layers (<2 segs)	reordered layers
LT Gruel VI	27 mm	338 666 mm	72 763 mm	2 206
RR Gruel VI	27 mm	139 313 mm	0 mm	612
LOuterGruel VI	27 mm	56 910 mm	0 mm	488

For `RR` and `LOuter`, **100 %** of remaining travel is inside reorderable layers; for `LT`, 82 %. So a stronger per-layer search had a clear target. Segment counts per layer are small — the maximum anywhere in the test set is 366, and the vast majority of layers hold 3–40 segments — so an additional O(n²) local search is affordable.

Implementation. `or_opt` pulls one segment out of the ordering and re-inserts it at the cheapest gap, in either orientation. It runs after 2-opt and is guarded by `or_opt_max_segments` (default 120) and `or_opt_max_moves` (default 150). It was added to `ORDERINGS` as `greedy+2opt+oropt` and to the `auto` candidate pool.

Bug found and fixed. The first run *increased* travel on `LT` (+1 638 mm) and `RT` (+1 105 mm) — impossible for a correct descent. Instrumenting a single layer isolated the fault: the insertion cost was computed as if inserting the segment in its *source* orientation (`rev == False` ⇒ entry = `seg.start`). When the segment was *currently reversed* in the ordering (`rev_i == True`), its current entry was actually `seg.end`, so the predicted entry/exit were swapped and the move was scored as an improvement when it was in fact a worsening. The fix is a one-token change — keep orientation relative to the segment's current direction:

```
c, f = (a_i, b_i) if rev == rev_i else (b_i, a_i)
```

After the fix the per-layer validator (`path_cost` before/after Or-opt) reported **0 worsening layers** and a net within-layer gain of 46 828 mm on `LT`. `two_opt` was audited too; it uses precomputed boundary arrays and is orientation-agnostic, so it was already correct.

Results (travel mm; full-set totals in §1):

File	Baseline	greedy+2opt	greedy+2opt+oropt	Δ vs 2-opt
Elevon Horn x 2	3 000.4	2 253.9	1 847.6	−18.0 %
Gruel Wing Plates	9 914.5	4 196.2	4 084.6	−2.7 %
LOuterGruel VI	154 557.9	56 936.6	56 958.2	+0.04 %
LR Gruel VI	443 449.0	145 831.1	142 877.9	−2.0 %
LT Gruel VI	871 749.7	411 455.6	411 059.5	−0.10 %
ROuterGruel VI	153 932.7	56 966.4	56 967.0	+0.001 %
RR Gruel VI	562 574.2	139 340.5	132 101.7	−5.2 %
RT Gruel VI	1 047 524.3	408 271.2	408 063.0	−0.05 %

Totals: `greedy+2opt+oropt` 1 213 959.5 (−62.61 % vs baseline), a further 11 292 mm (0.92 %) below 2-opt. `auto` scored 1 213 937.3 and selected Or-opt on six files, falling back to plain 2-opt on the two `Outer` files where Or-opt was a hair worse (auto's whole-file scoring catches this automatically).

Commentary. Or-opt is a clear win on layers with more segments (Elevon 18 %, RR 5.2 %, LR 2.0 %) and near-neutral on simple layers, where 2-opt already reaches the optimum. The time cost is modest (+3.1 s over the whole set). The larger lesson is the bug: a local search's move-evaluation must be proven non-worsening by an independent before/after check before its numbers are trusted — the first "improvement" numbers looked plausible yet were wrong.

Iteration 4 — remaining-travel accounting

After Or-opt, the same breakdown shows how little is left outside reachable layers (82–100 % of residual travel is in reordered layers), and the per-layer validator shows 2-opt+Or-opt is at/near the local optimum for typical 3–40-segment layers. The residual is therefore structural: whole-segment granularity and per-layer isolation (see §8), not a missing reorder opportunity.

Status: complete.

Iteration 5 — inter-layer coupling

Motivation. Every layer was ordered in isolation: the search minimised the path from the layer's entry but treated the layer's *exit* as free. Yet the connector into the next layer costs `distance(exit_prev, entry_next)`, so a layer that ends far from where the next one begins pays for it. The presentation idea that first suggested a fixed target turned out not to apply — `emit_segment` strips the prologue's XY travels and regenerates them — so both `entry_next` and `exit_prev` are free choices, and the objective is a chain. That calls for an alternating sweep rather than a one-pass lookahead.

Implementation. `two_opt` and `or_opt` gained an optional `end_target`; the path is then scored as *start* → … → *target*, and Or-opt is free to decide which segment finishes the layer. `_nn_build` was rebuilt around *sweeps*:

1. parse every chunk into units (fixed blocks vs reorderable layers) — untouched;
2. sweep 0 orders each layer with no target (the old behaviour);
3. each later sweep re-emits the file, passing every layer the *previous* sweep's entry point of the next layer as its target;
4. every sweep is scored with `analyze_lines` and the best text is kept.

Because sweep 0 is a candidate, coupling is **strictly non-worsening**. It is gated to the local-search orderings (`greedy+2opt`, `greedy+2opt+oropt`); `greedy` and `verbatim` have no exit-steering move, so coupling is skipped for them. Default `couple_layers = 1`.

Convergence check (travel mm, `greedy+2opt+oropt`):

File	couple=0	couple=1	couple=2	couple=3
LT Gruel VI	411 059.5	391 118.9	391 118.9	391 118.9
RT Gruel VI	408 063.0	396 972.0	396 972.0	396 972.0
RR Gruel VI	132 101.7	131 614.7	131 516.5	131 516.5

One sweep captures essentially all the gain; additional sweeps are near-neutral (`RR` gains a further 0.07 % at sweep 2). The extruding invariant held to < 0.006 mm.

Results. Full per-file comparison (`@0` = uncoupled):

File	Baseline	greedy+2opt	oropt @0	oropt coupled	Δ coupled vs @0
Elevon Horn x 2	3 000.4	2 253.9	1 847.6	1 847.6	0
Gruel Wing Plates	9 914.5	4 196.2	4 084.6	4 084.6	0
LOuterGruel VI	154 557.9	56 879.9	56 958.2	56 852.1	−106.1

File	Baseline	greedy+2opt	oropt @0	oropt coupled	Δ coupled vs @0
LR Gruel VI	443 449.0	145 831.1	142 877.9	142 877.9	0
LT Gruel VI	871 749.7	393 683.9	411 059.5	391 118.9	−19 940.6
ROuterGruel VI	153 932.7	56 883.5	56 967.0	56 875.1	−91.9
RR Gruel VI	562 574.2	138 933.6	132 101.7	131 614.7	−487.0
RT Gruel VI	1 047 524.3	399 102.1	408 063.0	396 972.0	−11 091.0

Totals: uncoupled `greedy+2opt+oropt` 1 213 959.5 (−62.61 %); coupled 1 182 242.9 (−63.59 %), a further **31 716.6 mm (−2.61 %)** removed. Coupling also lifts `greedy+2opt` to **1 197 764.2** (−63.11 %) — i.e. coupled 2-opt now beats uncoupled 2-opt+Or-opt.

Commentary. Coupling is the largest single addition since the greedy seed and the first to exploit structure *between* layers. It is cheap to reason about (non-worsening, best sweep kept) and, pleasingly, helps the weaker strategy (`greedy+2opt`) proportionally more, because that strategy benefits most from a well-chosen final edge. The two large `T` files dominate the gain; the dense `LR` and the small `Elevon`/`Wing` files were already well coupled, and `auto` selects the coupled result everywhere it helps.

Iteration 6 — (next)

Planned: segment splitting and/or a cheapest-insertion seed (§9).

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7. R/L symmetry analysis

7.1 Optimiser is mirror-invariant

Proven in Iteration 2: mirroring the input shifts every strategy's travel by the same constant. The optimiser contributes no asymmetry.

7.2 Are the input pairs actually mirrored?

An extruding bounding-box / Z-range scan of each pair:

Pair	width L/R	Y-height L/R	Z-range L/R	points L/R	Mirrored?
OuterGruel VI	160.193 / 160.192	238.798 / 238.798	0.20–96.60	56 268 / 56 487	yes
T Gruel VI	160.264 / 160.263	246.677 / 246.677	0.20–324.20	458 104 / 457 184	yes
R Gruel VI	158.941 / 158.629	249.887 / 249.772	0.20–97.40 / 0.20–115.80	279 149 / 312 267	**no**

`LR`/`RR` differ not only in point count but in height (Z range 97.40 vs 115.80) and layer count (520 vs 612) — they are **not** mirrored parts. `Outer` and `T` are close mirrors but not bit-identical (point counts differ by ~200–900, likely slicer tie-breaks and per-layer start-point conventions).

7.3 Measured L−R travel difference (mm; L minus R)

Pair	Baseline	verbatim	greedy	greedy+2opt	oropt @0	oropt coupled
OuterGruel VI	+625.2	+12.2	−348.1	−3.7	−8.8	−23.0
R Gruel VI	−119 125.2	−117 867.2	+13 659.9	+6 897.5	+10 776.2	+11 263.2

Pair	Baseline	verbatim	greedy	greedy+2opt	oropt @0	oropt coupled
T Gruel VI	−175 774.6	−181 893.5	−4 063.5	−5 418.2	+2 996.5	−5 853.1

Commentary. For the true mirrors (`Outer`, `T`) the optimiser collapses the gap from hundreds of thousands of mm (baseline) to a few thousand mm, and coupling takes `Outer` to −23.0 mm. The `T` figure moves with coupling (−5 853 mm; ~1.5 % of one file's travel) because the two inputs are close but not bit-identical mirrors, so the coupled sweep settles into slightly different local optima on each side. For `R Gruel` no strategy can equalise the pair, because the inputs are not mirrors (§7.2) — the optimiser is faithfully optimising two different shapes. The conclusion requested at project start — "L/R pairs should end up equal" — holds to the extent the inputs are mirrored; where they are not, the residual is a data issue, not an algorithm issue.

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8. Known remaining-travel causes

Measured after Or-opt + coupling. Over 82 % of the residual travel still sits in layers that *are* reordered, and the per-layer local search is at/near optimum for typical layers, so the remaining causes are structural:

- **Whole-segment granularity** — a long segment is treated as one unit; no splitting at an interior point to shorten a connector.
- **Mandatory non-XY moves** — Z lifts, retracts, and the start-up/trailing blocks (27 mm start-up is never reordered; layers with < 2 segments are verbatim).
- **Greedy seed ceiling** — the seed is typically 15–30 % above optimal, and the local searches only partially close that gap on large layers.
- **Large-layer guard** — layers above `or_opt_max_segments` (120) and `two_opt_max_segments` (800) skip the corresponding search; only a handful of layers in this set exceed them.
- **Coupling convergence** — one sweep captures essentially all the gain; a second sweep buys ≲ 0.1 %. The chain is solved to a good, not provably global, optimum.

Per-layer isolation was the largest listed cause in earlier drafts; it is now addressed by `couple_layers` (Iteration 5).

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9. Next steps

1. **Segment splitting** — allow a long segment to be split to shorten a connector (changes deposited geometry timing, so needs care). The headline remaining lever.
2. **Better seed** — cheapest-insertion or nearest-edge seed to lift the large-layer ceiling.
3. **Coupling follow-ups** — target the *entry* as well as the exit (currently only the exit is steered), or alternate forward/backward sweeps.
4. **Or-opt follow-ups** (cheap, likely small) — raise `or_opt_max_segments` or run a second Or-opt pass after a fresh 2-opt sweep.
5. Re-render this Markdown to PDF for the consolidated deliverable.